

Notes: Lazear (JPE, 2009)

Firm-Specific Human Capital: A Skill-Weights Approach

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1 Big picture

- **Question.** What exactly makes human capital firm-specific? Lazear asks whether firm-specificity must mean a skill that is literally useful only inside one firm, or whether specificity can arise because firms use ordinary general skills in different combinations.
- **Core idea.** The paper replaces the binary distinction between general and firm-specific skills with a **skill-weights** view. Every skill may be general, but each firm attaches different weights to those skills. A worker trained in a particular bundle may therefore be most productive at the original firm even though none of the component skills is unique to that firm.
- **Motivating example.** A software start-up may need managers who know programming, tax, and economics. Programming, tax, and economics are all general skills. The unusual combination is what is firm-specific.
- **Mechanism.** The worker invests in skills before period 2. The original firm and outside firms value those skills differently. Ex post wages are determined by bargaining between the worker and the firm, using outside opportunities as threat points.
- **Key implication 1: endogenous specificity.** Whether a skill bundle behaves like firm-specific capital depends on the thickness of the market. In thick markets, a worker can find another firm with similar weights, so apparent firm specificity disappears.
- **Key implication 2: firms may pay for general training.** A firm may finance training in skills that look general because the worker cannot fully capture their value outside the firm in the same combination.
- **Key implication 3: displacement losses.** Workers who are involuntarily separated lose wages when their skill bundle is poorly matched to outside firms. Losses are larger for workers from more idiosyncratic firms and in thinner labor markets.
- **Economic takeaway.** Firm-specific human capital can be understood as *general skills used in firm-specific proportions*. The paper therefore gives a microfoundation for tenure effects, displacement losses, firm-financed training, and the empirical importance of occupation and industry experience.

2 Incremental contribution of the paper

- **From labels to a mechanism.** Becker's framework classifies skills as general or firm-specific. Lazear gives a mechanism that endogenizes the classification: specificity arises from the match between a vector of general skills and firm-specific weights.
- **A bridge between human capital and matching.** The model is simultaneously a human-capital model and a matching model. Workers invest in a skill vector, and then match to firms whose weights make that vector valuable.
- **A new interpretation of tenure effects.** Wage losses after displacement need not come from knowledge that literally has zero value elsewhere. They can come from general skills that are not valued in the same proportions elsewhere.

- **A theory of firm payment for general training.** Traditional human-capital theory predicts that workers should pay for purely general training. The skill-weights view explains why firms may pay for skills that are general in isolation but specific in combination.
- **Market thickness becomes central.** Standard firm-specific capital theory does not naturally predict that wage losses should shrink in thick markets. In Lazear’s model, this follows directly: more outside draws increase the probability of finding a firm with similar weights.
- **Additional empirical content.** The model predicts how wage loss varies with market thickness, firm idiosyncrasy, layoff risk, occupation and industry controls, and the firm’s willingness to finance training.
- **One coherent framework.** Rather than explaining firm-financed training, displacement losses, occupational tenure effects, and matching with separate stories, the paper derives them from one skill-bundle model.

3 Model environment and notation

3.1 Skills and firm weights

There are two skills, A and B . The worker chooses how much of each skill to acquire during period 1. The cost function is

$$C(A, B), \quad C_A > 0, C_B > 0, C_{AA} > 0, C_{BB} > 0.$$

A firm i values the two skills according to a weight $\lambda_i \in [0, 1]$. Output at firm i is

$$Y_i(A, B) = \lambda_i A + (1 - \lambda_i)B.$$

Interpretation. If λ_i is high, firm i is A -intensive. If λ_i is low, firm i is B -intensive. No skill is intrinsically firm-specific. Firm specificity comes from the fact that the original firm may value A and B in a different mixture from most other firms.

3.2 Timing

1. The worker starts at an initial firm, indexed by 1, whose skill weight is λ_1 .
2. In period 1, the worker and firm make training investments A and B .
3. Before period 2, the worker receives an outside offer from another firm j with weight λ_j .
4. With exogenous probability q , the worker is laid off, for example because of a plant shutdown.
5. Period-2 wages are determined by Nash bargaining.

3.3 Period-2 bargaining wage

The worker’s productivity at the original firm is

$$Y_1 = \lambda_1 A + (1 - \lambda_1)B,$$

and productivity at the outside firm is

$$Y_j = \lambda_j A + (1 - \lambda_j)B.$$

With symmetric Nash bargaining, the period-2 wage is the average of the inside and outside values:

$$W_2 = \frac{1}{2}\{Y_1 + Y_j\} = B + \frac{1}{2}(\lambda_1 + \lambda_j)(A - B).$$

Interpretation. The worker's wage rises with outside opportunities even if he stays at the original firm, because the outside offer improves his threat point. This is why training that increases outside value affects bargaining and the split of rents.

3.4 Efficient separation condition

The worker stays with the initial firm if the initial firm values the skill bundle more than the outside firm:

$$Y_1 \geq Y_j.$$

Equivalently,

$$(\lambda_1 - \lambda_j)(A - B) \geq 0.$$

For an A -favoring worker, $A > B$, the stay condition becomes

$$\lambda_1 \geq \lambda_j.$$

If F is the distribution function of outside firm weights, then, ignoring exogenous layoffs, the probability of voluntary retention is $F(\lambda_1)$. Including exogenous layoff probability q , the probability of staying is

$$F(\lambda_1)(1 - q).$$

Interpretation. Turnover is endogenous to the match between the worker's skill bundle and firms' skill weights. A worker trained for a high- λ_1 firm is less likely to quit because few outside firms value A even more heavily.

4 Wage formulas and surplus

4.1 Expected wages for stayers, quitters, and laid-off workers

For $A > B$, the expected wage of a worker who stays is

$$W_{\text{stay}} = B + \frac{\lambda_1 + E(\lambda_j \mid \lambda_j < \lambda_1)}{2}(A - B).$$

The expected wage of a worker who quits is

$$W_{\text{quit}} = B + \frac{\lambda_1 + E(\lambda_j \mid \lambda_j > \lambda_1)}{2}(A - B).$$

If the worker is exogenously laid off, the paper assumes, for simplicity, that the outside job always beats leisure. The expected layoff wage is

$$W_{\text{layoff}} = \frac{1}{2} \left[\lambda_L A + (1 - \lambda_L) B + \bar{\lambda} A + (1 - \bar{\lambda}) B \right],$$

where λ_L is the weight attached to the worker's leisure fallback and $\bar{\lambda} = E(\lambda)$ is the mean outside firm weight.

4.2 Joint surplus of the worker and initial firm

The joint surplus maximized by the worker and original firm is

$$\begin{aligned} S(\lambda_1) = & F(\lambda_1)(1 - q) [\lambda_1 A + (1 - \lambda_1) B] \\ & + [1 - F(\lambda_1)](1 - q) E(W_{\text{quit}}) \\ & + q E(W_{\text{layoff}}) - C(A, B). \end{aligned}$$

Substituting the wage formulas gives the paper's expression

$$\begin{aligned} S(\lambda_1) = & F(\lambda_1)(1 - q) [\lambda_1 A + (1 - \lambda_1) B] \\ & + [1 - F(\lambda_1)](1 - q) \left[B + \frac{\lambda_1 + E(\lambda_j \mid \lambda_j > \lambda_1)}{2} (A - B) \right] \\ & + \frac{q}{2} [\lambda_L A + (1 - \lambda_L) B + \bar{\lambda} A + (1 - \bar{\lambda}) B] - C(A, B). \end{aligned}$$

Important qualification. This is the joint surplus of the worker and the initial firm, not full social surplus. If the worker quits or is laid off, part of output accrues to the new firm through bargaining. The initial worker-firm pair does not internalize that rent.

4.3 First-order conditions

The first-order condition for A is

$$\begin{aligned} C_A = & (1 - q) F(\lambda_1) \lambda_1 + (1 - q) [1 - F(\lambda_1)] \frac{\lambda_1 + E(\lambda_j \mid \lambda_j > \lambda_1)}{2} \\ & + \frac{q}{2} (\bar{\lambda} + \lambda_L). \end{aligned}$$

The first-order condition for B is

$$\begin{aligned} C_B = & (1 - q) F(\lambda_1) (1 - \lambda_1) \\ & + (1 - q) [1 - F(\lambda_1)] \left[1 - \frac{\lambda_1 + E(\lambda_j \mid \lambda_j > \lambda_1)}{2} \right] \\ & + \frac{q}{2} (1 - \bar{\lambda} + 1 - \lambda_L). \end{aligned}$$

Interpretation. The marginal benefit of each skill is a probability-weighted average of its value in three states: staying, quitting, and being laid off. The original firm's skill weights matter most when staying is likely; outside-market weights matter more when quitting or layoff is likely.

5 Main derivations and economic intuition

5.1 Why quits generate wage gains

Using the expected wage formulas,

$$W_{\text{quit}} - W_{\text{stay}} = \frac{1}{2} [E(\lambda_j \mid \lambda_j > \lambda_1) - E(\lambda_j \mid \lambda_j < \lambda_1)] (A - B).$$

For $A > B$, the bracketed term is positive. Therefore,

$$W_{\text{quit}} > W_{\text{stay}}.$$

Economic intuition. A voluntary quitter leaves only when an outside firm values the worker's skill bundle more highly. Therefore, quits are associated with wage gains. This is a matching implication, not a traditional firm-specific capital implication.

5.2 Why layoffs generate wage losses

The wage change after an involuntary layoff is

$$W_{\text{layoff}} - W_{\text{stay}} = \frac{1}{2} [\lambda_L + \bar{\lambda} - \lambda_1 - E(\lambda_j \mid \lambda_j < \lambda_1)] (A - B).$$

This expression is not mechanically signed. But for workers who voluntarily stay at the initial firm, the initial match is generally better than the expected outside match, so the expression is usually negative.

Economic intuition. A layoff breaks a good match. The worker is pushed into the outside market without choosing the best time or match. Wage loss is therefore the empirical shadow of an idiosyncratic skill bundle.

5.3 Idiosyncratic firms create more unbalanced investment

An idiosyncratic firm has an extreme λ_1 , for example λ_1 close to one for an A -intensive firm. Lazear shows that, for A -favoring firms, $A - B$ is maximized at $\lambda_1 = 1$.

The intuition follows from the A first-order condition. Raising λ_1 increases the value of A in the state in which the worker stays. Since high- λ_1 firms also retain workers with higher probability, the worker-firm pair chooses a more A -tilted investment strategy.

Implications.

- Workers at more idiosyncratic firms invest more unevenly across skills.
- Quit probabilities are lower: if $A > B$, a quit requires $\lambda_j > \lambda_1$, which becomes less likely as λ_1 rises.
- Layoff wage losses are larger: the worker's outside alternatives are less likely to value the same extreme skill bundle.

5.4 Low ex ante layoff risk increases idiosyncratic investment

If q is low, the worker expects to remain in the original firm. The marginal value of investing in the initial firm's favored skill rises. Therefore, workers in low-layoff-risk firms invest more heavily in the skill mix preferred by the initial firm.

Implications.

- Wages of stayers are higher when q is low.
- If a layoff occurs despite low ex ante layoff risk, the wage loss is larger because the worker invested more specifically in the initial firm's weights.
- More secure jobs can generate larger losses when unexpectedly destroyed.

5.5 Market thickness and endogenous specificity

A thicker market is modeled as more outside offers. Suppose the worker receives N independent draws of outside firm weights. For an A -favoring worker, the worker stays only if all N outside weights are below λ_1 , which occurs with probability

$$F(\lambda_1)^N.$$

With N outside offers, expected surplus becomes

$$S_N(\lambda_1) = F(\lambda_1)^N(1 - q)Y_1 + [1 - F(\lambda_1)^N](1 - q)E(W_{\text{quit}}) + qE(W_{\text{layoff}}) - C(A, B).$$

As $N \rightarrow \infty$, the worker can almost surely find a firm with the most favorable weight $\lambda_{\max}$. The highest and second-highest order statistics converge to the same value. Therefore, the wages of stayers, voluntary movers, and involuntary movers converge.

Core result. Even though investment becomes more idiosyncratic as the market thickens, wage loss from separation goes to zero in the limit.

Economic intuition. In a thin market, an unusual skill bundle is hard to redeploy. In a thick market, the same bundle becomes general because some other firm is likely to value it similarly. Thus, specificity is not a fixed technological property; it is a market-equilibrium property.

5.6 Why firms may pay for apparently general training

Suppose a firm wants a worker to learn a combination of tax, economics, and programming. Each skill is general alone, but the bundle may be unusual. A worker who leaves may find that outside firms use only part of the bundle. Therefore, the worker is unwilling to bear the entire training cost.

In the paper's numerical example, when $\gamma = 0.1$, $q = 0.05$, $\lambda_1 = 0.95$, $C(A, B) = (A^2 + B^2)/2$, and λ is uniform on $[0, 1]$, the solution has approximately

$$A = 0.93, \quad B = 0.07,$$

with total training cost around 0.43. The period-1 wage implies that the worker pays most of the cost, but the firm bears around 0.08, nearly one-fifth of the training cost.

Economic intuition. Firm payment for training does not prove that the skills are non-general in isolation. It may mean that the *combination* of general skills is not fully portable.

5.7 Worker choice, firm choice, and social efficiency

Appendix A compares three investment benchmarks.

Social optimum. Social surplus includes the full output of the worker wherever he works:

$$SS = F(\lambda_1)(1 - q)Y_1 + [1 - F(\lambda_1)](1 - q)E(Y_j \mid \lambda_j > \lambda_1) \\ + q[\bar{\lambda}A + (1 - \bar{\lambda})B] - C(A, B).$$

The social planner internalizes the new firm's surplus when the worker leaves.

Joint worker-initial-firm choice. The pair underweights the new firm's surplus. Relative to the social optimum, this can generate underinvestment in the skill most valued by future outside firms and overinvestment in the skill that improves the current bargaining position.

Worker pays for training. If the worker chooses and bears the training cost, he invests too little in the skill that favors the current firm and too much in the outside-option skill, because outside value improves his bargaining position.

Firm pays for training. If the firm chooses and pays, the firm underinvests in both skills because training raises the worker's bargaining power. The firm also biases training toward the skill favored by the current firm and away from portability.

Takeaway. Training investment is efficient only under strong contractibility assumptions. Once investment choice and cost incidence are separated, bargaining incentives distort both the level and the composition of training.

6 Empirical implications and how to test the model

6.1 Prediction 1: market thickness reduces displacement losses

Wage losses after involuntary turnover should be smaller in thick labor markets. Possible proxies include:

- number of relevant vacancies per worker,
- local population density,
- number of nearby firms in the same industry or occupation,
- occupational concentration or diversification in the local labor market.

Distinguishing feature. Traditional firm-specific capital theory does not naturally predict that the same skill becomes less firm-specific when the market thickens. The skill-weights view does.

6.2 Prediction 2: idiosyncratic firms have lower quits and larger layoff losses

Firms with unusual skill mixes should display:

- lower quit rates,
- more unbalanced training profiles,
- larger wage losses after involuntary separation,

- higher wages for stayers if the firm-specific bundle is very valuable internally.

Measurement is difficult, but proxies include unusual occupational mixes within a firm, unusual combinations of skills in job postings, young or emerging industries, and firms combining technical fields that are rarely combined elsewhere.

6.3 Prediction 3: firms finance general training when bundles are nonportable

A firm may pay for training in skills that are general in isolation if the specific combination is valuable mainly inside the firm. Empirically, firm-financed training should be more common when:

- the bundle of skills is unusual,
- the firm is idiosyncratic,
- outside matching opportunities are thin,
- the worker faces a large expected wage loss after displacement.

6.4 Prediction 4: occupation and industry controls reduce tenure effects

If occupations and industries proxy for skill weights, then controlling for occupational and industry experience should reduce the estimated pure firm-tenure effect in wage regressions. This is why empirical work finding strong industry or occupation tenure effects supports the skill-weights interpretation.

6.5 Prediction 5: layoff risk affects training composition

Workers in firms with low ex ante layoff risk should invest more heavily in the firm’s favored skill mix. Therefore:

- stayers in low-layoff-risk firms should have higher wages,
- conditional on an unexpected layoff, those workers should experience larger wage losses,
- high-layoff-risk environments should induce more balanced, portable skill investment.

6.6 Prediction 6: business cycles and mass layoffs

In recessions, markets become thinner: more workers search and fewer vacancies exist. Therefore, displacement losses should be larger in downturns. Mass layoffs can also thin the relevant market because many similarly trained workers search simultaneously. However, this effect may interact with stigma effects: mass layoffs may reduce negative inference about individual productivity while increasing competition for similar jobs.

7 Tables and extracted panels from the paper

The paper is primarily theoretical and contains only one formal table and no conventional figures. The notes therefore extract the central equation panels and the appendix table, because these are

the closest visual objects in the paper and they summarize the model's core mechanics.

7.1 Model setup, surplus, and investment conditions

Read:

- The first panel presents the core output function and the Nash bargaining wage.
- The second panel shows the expected joint surplus expression that combines stay, quit, and layoff states.
- The surplus expression is the foundation for the model's investment predictions.

Economic message: The worker's skills are general components, but their value depends on the firm's weight vector. The worker and firm invest not just for the initial match, but for a probability-weighted portfolio of future matching outcomes.

skills that he takes into that period. The worker's output at firm j is given by

$$\lambda_j A + (1 - \lambda_j) B.$$

Because the situation is one of ex post bilateral monopoly, the second-period wage is determined by the equilibrium to some bargaining game. One extreme is to give all the ex post rents to the firm, paying the worker no more than the alternative wage offer. The other extreme is to give all the ex post rents to the worker, paying exactly the full value of output. All investment increases the alternative wage as well, although not necessarily by the same amount elsewhere as in the current firm.

For sake of conformity with much of the economic literature, it is assumed that the wage during period 2 is determined according to a Nash bargaining framework. Since the worker is worth $\lambda_j A + (1 - \lambda_j) B$ at the alternative firm, the alternative firm would be willing to pay up to $\lambda_j A + (1 - \lambda_j) B$ for the worker's services. The initial firm receives output of $\lambda_1 A + (1 - \lambda_1) B$, so the Nash bargain implies that the wage in period 2 is

$$W_2 = \frac{1}{2}[\lambda_1 A + (1 - \lambda_1) B] + [\lambda_j A + (1 - \lambda_j) B] \quad (1)$$

or

$$W_2 = B + \frac{1}{2}(\lambda_1 + \lambda_j)(A - B).$$

(a) Model output and period-2 wage.

The investment problem is to choose A and B , knowing that the worker may remain at the initial firm, but that there is some chance that he will move to another firm, either because he gets a better outside offer or because his firm shuts down. The privately efficient choice of A and B maximizes expected net surplus to the combination of the worker and firm:⁵

$$S(\lambda_1) = F(\lambda_1)(1 - q)[\lambda_1 A + (1 - \lambda_1) B] + (1 - q)[1 - F(\lambda_1)]E(W_{\text{quit}}) + qE(W_{\text{layoff}}) - C(A, B)$$

or

$$S(\lambda_1) = F(\lambda_1)(1 - q)[\lambda_1 A + (1 - \lambda_1) B] + [1 - F(\lambda_1)](1 - q) \left[B + \frac{[\lambda_1 + E(\lambda_j | \lambda_j > \lambda_1)](A - B)}{2} \right] + \frac{q}{2}[\lambda_1 A + (1 - \lambda_1) B + \bar{\lambda} A + (1 - \bar{\lambda}) B] - C(A, B). \quad (2)$$

Note that $S(\lambda_1)$ is firm and worker joint surplus. It is not the socially efficient amount because the second and third terms of the expression ignore the surplus that goes to other firms through the bargaining process in those situations in which the worker leaves the firm.⁶ The

(b) Expected joint surplus.

7.2 First-order conditions and cost sharing

Read:

- The first-order conditions decompose the marginal value of each skill into stay, quit, and layoff terms.
- The period-1 wage expression shows how the training cost and rent split are built into the initial wage.
- The first-period wage is the accounting device that determines who effectively pays for training.

Economic message: Training cost incidence is not pinned down only by whether skills are general or specific. It also depends on bargaining, outside opportunities, and the probability that the worker remains with the initial firm.

earnings with his current firm, which happens $(\lambda_1)(1 - q)$ of the time. The second term reflects earnings when he receives a better outside offer than at the current firm and his current firm does not shut down. This occurs whenever $\lambda_j > \lambda_1$ and when no layoff occurs, or $[1 - F(\lambda_1)](1 - q)$ of the time.⁷ Finally, the firm may shut down, in which case he must take the alternative job with earnings shown in the third term. This occurs q of the time. The last two terms ignore the part of output that is captured by other firms.

The first-order conditions are

$$\begin{aligned} \frac{\partial}{\partial A} = & (1 - q)F(\lambda_1)\lambda_1 + (1 - q)[1 - F(\lambda_1)]\left[\frac{\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)}{2}\right] \\ & + \frac{q}{2}(\bar{\lambda} + \lambda_L) - C_A = 0 \end{aligned} \quad (3)$$

and

$$\begin{aligned} \frac{\partial}{\partial B} = & (1 - q)F(\lambda_1)(1 - \lambda_1) \\ & + (1 - q)[1 - F(\lambda_1)]\left[1 - \frac{\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)}{2}\right] \\ & + \frac{q}{2}(1 - \bar{\lambda} + 1 - \lambda_L) - C_B = 0. \end{aligned} \quad (4)$$

(a) Investment first-order conditions.

$$\begin{aligned} & (\lambda_1 A + (1 - \lambda_1)B - \{B + \frac{1}{2}[\lambda_1 + E(\lambda_j|\lambda_j < \lambda_1)]\})(1 - q)F(\lambda_1) - \\ & W_1 - C(A, B), \end{aligned}$$

and this must equal $\gamma S(\lambda_1)$. After the appropriate substitutions, one obtains

$$\begin{aligned} W_1 = & (\text{Output if Stay})(1 - \gamma)(1 - q)F(\lambda_1) - W_{\text{stay}}(1 - q)F(\lambda_1) \\ & - (W_{\text{quit}})(1 - q)[1 - F(\lambda_1)] - W_{\text{layoff}}q - (1 - \gamma)C(A, B) \end{aligned}$$

or

$$\begin{aligned} W_1 = & [\lambda_1 A + (1 - \lambda_1)B](1 - \gamma)(1 - q)F(\lambda_1) \\ & - \left[B + \frac{\lambda_1 + E(\lambda_j|\lambda_j < \lambda_1)}{2}(A - B)\right](1 - q)F(\lambda_1) \\ & - \left[B + \frac{\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)}{2}(A - B)\right](1 - q)[1 - F(\lambda_1)] \\ & - \frac{q}{2}[\lambda_L A + (1 - \lambda_L)B + \bar{\lambda} A + (1 - \bar{\lambda})B] - (1 - \gamma)C(A, B). \end{aligned} \quad (5)$$

The first-period wage is the rent-splitting parameter, which is determined taking into account that the second-period wage will be set by a

(b) Period-1 wage and training cost split.

7.3 Wage changes and market thickness

Read:

- The wage-change equations separate voluntary quits from involuntary layoffs.
- Voluntary quitters gain because they move to firms with better weights.
- The market-thickness panel introduces N outside offers and shows how thicker markets change the separation problem.

Economic message: The same human capital can look firm-specific in thin markets and general in thick markets. This is the paper's most distinctive departure from the standard view.

$$W_{\text{quit}} - W_{\text{stay}} = \frac{1}{2}[E(\lambda_j|\lambda_j > \lambda_1) - E(\lambda_j|\lambda_j < \lambda_1)](A - B). \quad (6)$$

The expression in (6) is positive. Workers quit only when they do better elsewhere.

Similarly, the expected wage change among involuntary leavers is

$$W_{\text{layoff}} - W_{\text{stay}} = \frac{1}{2}[\lambda_L + \bar{\lambda} - \lambda_1 - E(\lambda_j|\lambda_j < \lambda_1)](A - B). \quad (7)$$

Although one would expect (7) to be negative since stayers should do better than those who get laid off, the expression cannot be signed.¹⁰ But were $W_{\text{layoff}} > W_{\text{stay}}$, the firm would only be in the training business. No worker would choose to stay voluntarily during the second period since he does better by taking his chances on the outside market.

Idiosyncratic Firms and Unbalanced Investment

The pattern of investment is more unbalanced for workers whose initial job is at a firm with relatively idiosyncratic weights. This is not surprising since a primary reason for investing in skills is to use them in the current firm.

(a) Wage changes and idiosyncratic firms.

that would prevail in the absence of any involuntary mobility.

An increase in market thickness is modeled here as allowing more offers. The model above assumes that before the work period begins, the worker gets a new draw of λ . A thicker market is represented as one in which the worker gets multiple independent draws of λ before the work period and can select the job that best suits his prior investment strategy. There are two key results in this section. First, as the market gets extremely thick, investment becomes idiosyncratic; that is, $A - B$ increases. Second, and despite this, as the market thickens, the wage loss associated with separation declines.

The probability that the current weight λ_1 exceeds all N draws is $F(\lambda_1)^N$, so expected surplus in (2) is now given by

$$\begin{aligned} S(\lambda_1) = & F(\lambda_1)^N(1 - q)[\lambda_1 A + (1 - \lambda_1)B] \\ & + [1 - F(\lambda_1)^N](1 - q)E(W_{\text{quit}}) \\ & + qE(W_{\text{layoff}}) - C(A, B). \end{aligned} \quad (2')$$

As is intuitive, expected joint surplus, (2'), increases in thickness of the market. The more offers the better even though this results in a higher probability of separation from the original firm. Formally, first

¹¹ Using (3') in App. B, note that the derivative of the left-hand side with respect to q is

(b) Market thickness and outside offers.

7.4 Social efficiency and investment distortions

Read:

- Appendix A compares the socially efficient investment choice with the worker-initial-firm joint surplus choice.
- The later panel compares worker-financed and firm-financed training incentives.
- The equations show that the identity of the party choosing and paying for training changes the direction of investment distortions.

Economic message: Even if workers and firms bargain efficiently ex post, training can be distorted ex ante because some returns accrue to future firms or because training changes bargaining power.

maximize the joint surplus in (2). The socially efficient level of surplus takes into account the rents going to other firms in the economy when a worker quits or is laid off. The joint surplus in (2) ignores the rent to other firms and focuses only on that going to the worker or current firm.

The social surplus is then

$$\begin{aligned} \text{Social Surplus} = & F(\lambda_1)(1-q)[\lambda_1 A + (1-\lambda_1)B] \\ & + [1-F(\lambda_1)](1-q)[E(\lambda_j|\lambda_j > \lambda_1)A \\ & + [1-E(\lambda_j|\lambda_j > \lambda_1)]B] \\ & + q[\bar{\lambda}A + (1-\bar{\lambda})B] - C(A, B). \end{aligned} \quad (2A)$$

The first-order conditions are

$$\frac{\partial}{\partial A} = (1-q)F(\lambda_1)\lambda_1 + (1-q)[1-F(\lambda_1)]E(\lambda_j|\lambda_j > \lambda_1) + q\bar{\lambda} - C_A = 0 \quad (3A)$$

and

$$\begin{aligned} \frac{\partial}{\partial B} = & (1-q)F(\lambda_1)(1-\lambda_1) + (1-q)[1-F(\lambda_1)][1-E(\lambda_j|\lambda_j > \lambda_1)] \\ & + q(1-\bar{\lambda}) - C_B = 0. \end{aligned} \quad (4A)$$

Recall that the first-order conditions for joint surplus maximization are

$$\begin{aligned} \frac{\partial}{\partial A} = & (1-q)F(\lambda_1)\lambda_1 + (1-q)[1-F(\lambda_1)]\left[\frac{\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)}{2}\right] \\ & + \frac{q}{2}(\bar{\lambda} + \lambda_t) - C_A = 0 \end{aligned} \quad (3)$$

and

$$\frac{\partial}{\partial B} = (1-q)F(\lambda_1)(1-\lambda_1) + (1-q)[1-F(\lambda_1)]\left[1 - \frac{\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)}{2}\right]$$

(a) Social surplus versus joint surplus.

Were workers allowed to choose A and B and were they required to bear the costs of investment, the maximization problem would be

$$\begin{aligned} & F(\lambda_1)(1-q)\left[B + \frac{[\lambda_1 + E(\lambda_j|\lambda_j \leq \lambda_1)](A-B)}{2}\right] \\ & + [1-F(\lambda_1)](1-q)\left[B + \frac{[\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)](A-B)}{2}\right] \\ & + \frac{q}{2}[\lambda_t A + (1-\lambda_t)B + \bar{\lambda}A + (1-\bar{\lambda})B] - C(A, B) \end{aligned} \quad (2A')$$

with first-order conditions

$$\begin{aligned} \frac{\partial}{\partial A} = & (1-q)F(\lambda_1)\left[\frac{\lambda_1 + E(\lambda_j|\lambda_j \leq \lambda_1)}{2}\right] \\ & + (1-q)[1-F(\lambda_1)]\left[\frac{\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)}{2}\right] \\ & + \frac{q}{2}(\bar{\lambda} + \lambda_t) - C_A = 0 \end{aligned} \quad (3A')$$

and

$$\begin{aligned} \frac{\partial}{\partial B} = & (1-q)F(\lambda_1)\left[1 - \frac{\lambda_1 + E(\lambda_j|\lambda_j \leq \lambda_1)}{2}\right] \\ & + (1-q)[1-F(\lambda_1)]\left[1 - \frac{\lambda_1 + E(\lambda_j|\lambda_j > \lambda_1)}{2}\right] \\ & + \frac{q}{2}(1-\bar{\lambda} + 1-\lambda_t) - C_B = 0 \end{aligned} \quad (4A')$$

Except for the first-terms, (3A') and (4A') are identical to (3) and (4). It is clear from the comparison of the first terms that the worker would choose too little A and too much B . This improves the worker's net return when he stays at the firm because he stays when $\lambda_j < \lambda_1$. The worker overinvests in outside

(b) Worker-choice and firm-choice investment incentives.

7.5 Table D1: numerical illustration of idiosyncratic weights

Read:

- Table D1 reports how the optimal skill mix changes as λ_1 increases.
- As the firm becomes more A -intensive, optimal A rises and optimal B falls.
- The period-1 wage is not monotone in λ_1 in this numerical example.

Economic message: The table makes the model's comparative statics concrete: idiosyncratic firm weights tilt the worker's skill investment toward the firm's favored skill, even when the component

skills are general.

TABLE D1
EFFECT OF λ_1 ON A , B , AND W_1

λ_1	A	B	W_1
.5	.56	.44	-.51
.75	.75	.25	-.49
.95	.93	.07	-.50
1	.98	.03	-.50

Table D1: effect of λ_1 on A , B , and W_1 .

8 Short summary

- Lazear argues that most firm-specific human capital can be viewed as general skills combined in firm-specific weights.
- Output is $\lambda_i A + (1 - \lambda_i)B$, so firms differ in how they value the same skill vector.
- The worker invests in A and B before knowing whether he will stay, quit, or be laid off.
- Nash bargaining makes wages depend on both current productivity and outside opportunities.
- Voluntary quits generate wage gains because quitters move to firms with more favorable weights.
- Involuntary layoffs often generate wage losses because a good match is destroyed.
- More idiosyncratic firms induce more unbalanced investment and lower quit rates.
- Thicker labor markets reduce wage loss from displacement because similar matches are easier to find.
- Firms may pay for general training when the skill bundle is not fully portable.
- Industry and occupation experience can proxy for skill weights and reduce estimated pure firm-tenure effects.

9 Conclusion

9.1 Core conclusion

The paper gives a powerful reinterpretation of firm-specific human capital. Skills need not be literally useful only at one firm. A skill is effectively firm-specific when its value depends on a combination of weights that few other firms share.

9.2 Economic intuition

A worker’s skill vector is like a portfolio. Each firm prices the components of that portfolio differently. A worker trained for a firm with unusual weights may suffer a loss when separated because other firms do not value the portfolio in the same proportions. Conversely, when the market is thick, another similar buyer for the skill portfolio can be found, and the same training becomes effectively general.

9.3 Incremental contribution revisited

The paper’s main contribution is not merely to rename firm-specific capital. It makes specificity endogenous to the distribution of firms and the thickness of the market. This yields new predictions about displacement losses, quit behavior, firm-financed training, occupational tenure, and business-cycle variation in wage losses.

9.4 Implications for empirical work

The paper suggests that wage regressions should pay close attention to local market thickness, industry and occupation experience, unusual skill bundles, and ex ante layoff risk. Pure firm tenure may overstate traditional firm-specific human capital if it is actually capturing occupation- or industry-specific skill weights.

9.5 Limitations

The model is deliberately stylized. It uses two skills, symmetric Nash bargaining, observable investment, and a simple separation structure. Actual labor markets have many skills, search frictions, asymmetric information, training contracts, and firm-specific personnel practices. Still, the two-skill model is useful because it isolates a clean mechanism.

9.6 Takeaway

Firm-specific human capital can be the product of general skills placed in a firm-specific recipe. The portability of a worker’s human capital depends not only on what the worker knows, but also on how many other firms want that same combination.